

29-3d Bank Capital, Leverage, and the Financial Crisis of 2008–2009

In the previous sections, we have seen a simplified explanation of how banks work. But the reality of modern banking is a bit more complex, and this complexity played a key role in the financial crisis of 2008 and 2009. Before looking at that crisis, we need to learn a bit more about how banks actually function.

In each of the bank balance sheets we have seen so far, a bank accepts deposits and either uses those deposits to make loans or holds them as reserves. More realistically, a bank gets financial resources not only from accepting deposits but also, like other companies, from issuing equity and debt. The resources that a bank obtains from issuing equity to its owners are called bank capital (the resources a bank’s owners have put into the institution) . A bank uses these financial resources in various ways to generate profit for its owners. It not only makes loans and holds reserves but also buys financial securities, such as stocks and bonds.

Here is a more realistic example of a bank’s balance sheet:

More Realistic National Bank			
Assets		Liabilities and Owners’ Equity	
Reserves	\$200	Deposits	\$800
Loans	700	Debt	150
Securities	100	Capital (owners’ equity)	50

On the right side of this balance sheet are the bank’s liabilities and capital (also called *owners’ equity*). This bank obtained \$50 of resources from its owners. It also took in \$800 of deposits and issued \$150 of debt. The total of \$1,000 was put to use in three ways, listed on the left side of the balance sheet, which shows the bank’s assets. This bank held \$200 in reserves, made \$700 in bank loans, and used \$100 to buy financial securities, such as government or corporate bonds. The bank decides how to allocate its resources among asset classes based on their risk and return, as well as on any regulations (such as reserve requirements) that restrict the bank’s choices.

By the rules of accounting, the reserves, loans, and securities on the left side of the balance sheet must always equal, in total, the deposits, debt, and capital on the right side of the balance sheet. There is no magic in this equality. It occurs because the value of the owners’

equity is, by definition, the value of the bank's assets (reserves, loans, and securities) minus the value of its liabilities (deposits and debt). Therefore, the left and right sides of the balance sheet always sum to the same total.

Many businesses rely on

leverage (the use of borrowed money to supplement existing funds for purposes of investment)

, the use of borrowed money to supplement existing funds for investment purposes. Indeed, whenever a business uses debt to finance an investment project, it is applying leverage. Leverage is particularly important for banks, however, because borrowing and lending are at the heart of what they do. To fully understand banking, therefore, it is crucial to understand how leverage works.

The leverage ratio (the ratio of assets to bank capital) is the ratio of the bank's total assets to bank capital. In this example, the leverage ratio is $\$1,000/\50 , or 20. A leverage ratio of 20 means that for every dollar of capital that the bank owners have contributed, the bank has \$20 of assets. Of the \$20 of assets, \$19 are financed with borrowed money—either by taking in deposits or issuing debt.

You may have learned in a science class that a lever can amplify a force: A boulder that you cannot move with your arms alone will move more easily if you use a lever. A similar result occurs with bank leverage. To see how this amplification works, let's continue with this numerical example. Suppose that the bank's assets were to rise in value by 5 percent because, say, some of the securities the bank was holding rose in price. Then the \$1,000 of assets would now be worth \$1,050. Because the depositors and debt holders are still owed \$950, the bank capital rises from \$50 to \$100 ($\$1050 - \950). Thus, when the leverage rate is 20, a 5-percent increase in the value of assets increases the owners' equity by 100 percent.

The same principle works on the downside, but with troubling consequences. Suppose that some people who borrowed from the bank default on their loans, reducing the value of the bank's assets by 5 percent, to \$950. Once again, the depositors and debt holders are still owed \$950, so the value of the owners' equity falls to zero ($\$950 - \950). Thus, when the leverage ratio is 20, a 5-percent fall in the value of the bank's assets leads to a 100-percent fall in bank capital. If the value of assets were to fall by more than 5 percent, the bank's assets would fall below its liabilities. In this case, the bank would be *insolvent*, and it would be unable to pay off its debt holders and depositors in full.

Bank regulators require banks to hold a certain amount of capital. The goal of such a capital requirement (a government regulation specifying a minimum amount of bank capital)

is to ensure that banks will be able to pay off their depositors (without having to resort to government-provided deposit insurance funds). The amount of capital required depends on the kind of assets a bank holds. A bank holding risky assets such as loans to borrowers whose credit is of dubious quality would be, all other things equal, required to hold more capital than a bank holding safe assets such as government bonds.

Economic turmoil can result when banks find themselves with too little capital to satisfy capital requirements. An example of this phenomenon arose in 2007 and 2008, when many banks incurred sizable losses on some of their assets—specifically, mortgage loans and securities backed by mortgage loans. The shortage of capital induced the banks to reduce lending, a phenomenon called a *credit crunch*, which in turn contributed to a severe downturn in economic activity. (This event is discussed more fully in [Chapter 33](#).) To address this problem, the U.S. Treasury, working together with the Fed, put many billions of dollars of public funds into the banking system to increase the amount of bank capital. As a result, it temporarily made the U.S. taxpayer a part owner of many banks. The goal of this unusual policy was to recapitalize the banking system so that bank lending could return to a more normal level. And, in fact, by late 2009, it did.

QuickQuiz

5. Isabella takes **\$100** of currency from her wallet and deposits it into her checking account. If the bank adds the entire **\$100** to reserves, the money supply _____, but if the bank lends out some of the **\$100**, the money supply _____.
- a. increases; increases even more
 - b. increases; increases by less
 - c. is unchanged; increases
 - d. decreases; decreases by less
6. If the reserve ratio is $\frac{1}{4}$ and the central bank increases the quantity of reserves in the banking system by **\$120**, the money supply increases by
- a. **\$90**.
 - b. **\$150**.
 - c. **\$160**.
 - d. **\$480**.
7. A bank has capital of **\$200** and a leverage ratio of **5**. If the value of the bank's assets declines by **10** percent, then its capital will be reduced to
- a. **\$100**.
 - b. **\$150**.

c. \$180.

d. \$185.

Chapter 29: The Monetary System: 29-3d Bank Capital, Leverage, and the Financial Crisis of 2008–2009

Book Title: WGU Economics

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